KULIMA AFRICA

SAMPLE · PILOT DEMONSTRATION · NOT A FINANCING APPROVAL

# Demand-Signal Prospectus

Verified Coordination Patterns for Infrastructure Planning

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| DOCUMENT | KULIMA OS — Demand-Signal Prospectus (Sample) |
| REGION | Pilot Region — Rural Energy Planning |
| PERIOD | 7-cycle observation window (Week 1) |
| DATE GENERATED | 2026-05-05 |
| SYSTEM | KULIMA OS Pilot v0.2 — LUMOZA + ZENTARI + Critical Load Protection |
| CLASSIFICATION | Institutional Planning Artifact — Not for distribution as marketing |

Prepared by Kulima Africa for utilities, development finance institutions, and policymakers.
This document presents verified coordination patterns to inform infrastructure planning decisions.
It contains no personal data and is not a financing approval.

# 1 Executive Overview

Across rural and emerging economies, productive electricity demand — milling, irrigation, cold storage, water systems, clinics, and schools — is largely **invisible** to utilities and financiers. Infrastructure is therefore sized against household consumption proxies, leaving productive activity unserved and infrastructure investment without a credible demand basis.

This **Demand-Signal Prospectus** presents verified coordination patterns observed by KULIMA OS during a seven-cycle pilot window. It is intended to support utility planning, transformer sizing, capacity allocation, and institutional investment review.

### What this document enables

- Capacity and transformer sizing grounded in observed coordination, not assumed demand.
- Prioritisation of high-confidence zones for staged infrastructure deployment.
- Architectural protection of essential services (clinics, schools, water, emergency).
- A bankable, conservative lower-bound load basis for development finance review.

### What this document does *not* do

- It does **not** contain personal data, household identifiers, or individual profiles.
- It is **not** a credit assessment, loan approval, or financing instrument.
- It does **not** replace engineering design, ESIA, or community consultation.
- It does **not** guarantee future demand; coordination patterns require ongoing monitoring.

### Key pilot facts

| | |
|---|---|
| **Observation window** | 7 cycles (Week 1) |
| **Method** | Coordination-pattern detection from identity-free signals, telemetry-validated |
| **Privacy posture** | Zero-PII; raw signals never stored or exported |
| **Patterns detected** | 10 stable patterns across 3 zones |
| **High-confidence patterns** | 2 (suitable for infrastructure commitment) |
| **Essential services protected** | 5 (clinic, school, water systems, emergency services) |
| **Critical load reservation** | 25% baseline, architecturally enforced |

# 2 Coordination Summary

The pilot detected ten stable coordination patterns across three zones. Two patterns met the high-confidence threshold required for direct infrastructure commitment; one further pattern reached moderate confidence; the remainder are essential-service loads validated by telemetry or productive patterns awaiting strengthening.

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|---|---|
| **Zones with coordinated demand** | zone_a, zone_b, zone_c |
| **Total coordination patterns** | 10 |
| **High-confidence patterns** | 2 |
| **Moderate-confidence patterns** | 1 |
| **Low / insufficient productive patterns** | 2 (monitor only) |
| **Essential-service patterns (telemetry-validated)** | 5 |
| **Productive activities detected** | Cold storage, irrigation, milling |
| **Essential services detected** | Clinic, emergency services, school, water system |

### Confidence distribution

| Confidence band | Count | Share | Planning treatment |
|---|---|---|---|
| High (≥ 0.70) | 2 | 20% | Suitable for infrastructure commitment |
| Moderate (0.50–0.70) | 1 | 10% | Plan with contingency; reassess each cycle |
| Low (< 0.50) | 1 | 10% | Insufficient basis for commitment |
| Essential services (telemetry) | 5 | 50% | Non-negotiable, architecturally protected |
| Insufficient | 1 | 10% | Monitor only |

# 3 Verified Coordination Patterns

The following tables summarise verified coordination patterns, grouped by planning category. Each row reports the activity, zone, time window, observed stability across the seven-cycle window, coordination confidence, and the planning implication. Essential services are recorded as *N/A* for confidence because they are protected loads, not optimisation candidates.

### 3.1 — High-confidence productive patterns

| Activity | Zone | Time window | Stability | Confidence | Planning implication |
|---|---|---|---|---|---|
| Irrigation | zone_a | Morning | Stable (6/7) | 0.86 — High | Reliable morning power for water pumping |
| Cold storage | zone_b | Evening | Stable (6/7) | 0.86 — High | Evening capacity for cold-storage operations |

### 3.2 — Moderate-confidence productive patterns

| Activity | Zone | Time window | Stability | Confidence | Planning implication |
|---|---|---|---|---|---|
| Milling | zone_a | Afternoon | Stable (5/7) | 0.71 — Moderate | Afternoon capacity for grain processing |

### 3.3 — Low / insufficient productive patterns (monitor only)

| Activity | Zone | Time window | Stability | Confidence | Planning implication |
|---|---|---|---|---|---|
| Milling | zone_c | Morning | Intermediate (4/7) | 0.48 — Low | Monitor before infrastructure commitment |
| Irrigation | zone_c | Afternoon | Intermediate (3/7) | 0.26 — Insufficient | Insufficient basis for planning |

### 3.4 — Essential services (telemetry-validated, protected)

| Activity | Zone | Time window | Stability | Confidence | Planning implication |
|---|---|---|---|---|---|
| Clinic | zone_a | Continuous | Stable (7/7) | N/A — Essential | CRITICAL — non-negotiable priority load |
| School | zone_a | Morning | Stable (5/7) | N/A — Essential | CRITICAL — non-negotiable priority load |
| Water system | zone_b | Morning | Stable (7/7) | N/A — Essential | CRITICAL — non-negotiable priority load |
| Emergency services | zone_b | Continuous | Stable (7/7) | N/A — Essential | CRITICAL — non-negotiable priority load |
| Water system | zone_c | Morning | Stable (6/7) | N/A — Essential | CRITICAL — non-negotiable priority load |

*Validation note:* Productive patterns are corroborated by infrastructure telemetry where available; essential services are confirmed by telemetry only. Stability is reported as the number of cycles in which the pattern recurred within the seven-cycle observation window.

# 4 Critical Load Protection Policy

**Non-negotiable, architecturally enforced.** A 25% baseline of available capacity is permanently reserved for essential services and is excluded from optimisation, monetisation, and load-shedding logic. This reservation cannot be overridden operationally.

### Scenario reservation thresholds

| Scenario | Description | Essential load | Available for productive use |
|---|---|---|---|
| Baseline | Normal operation, all essential services active | 25% | 75% |
| Peak | Peak demand with simultaneous essential operation | 35% | 65% |
| Shock | Emergency requiring maximum essential capacity | 50% | 50% |

### Planning requirements

- Infrastructure must reserve 25% baseline capacity for essential services.
- Essential service loads cannot be shed during peak-demand periods.
- Productive-use optimisation operates within remaining capacity only.
- Design must accommodate 35% essential load under normal peak conditions.
- Design must permit scaling essential capacity to 50% in shock scenarios.

### Protected (non-negotiable) loads

| Service | Zone | Time window | Stability | Priority |
|---|---|---|---|---|
| Clinic | zone_a | Continuous | Stable | CRITICAL |
| School | zone_a | Morning | Stable | CRITICAL |
| Water system | zone_b | Morning | Stable | CRITICAL |
| Emergency services | zone_b | Continuous | Stable | CRITICAL |
| Water system | zone_c | Morning | Stable | CRITICAL |

# 5 Load Estimation & Capacity Guidance

Load estimates use **conservative lower-bound** activity-level profiles drawn from the World Bank Rural Electrification Toolkit (2008), ESMAP Technical Papers 121, 145 and 156, the IFC Productive Use of Energy Study (2018), and WHO Health Facility Electrification Guidelines (2020). Diversity and load factors are applied to reflect non-simultaneous operation.

### Total system demand (diversified)

| | |
|---|---|
| **Peak demand** | 15.66 kW |
| **Daily energy** | 68.35 kWh |
| **Monthly energy** | 2,050.5 kWh |
| **Annual energy** | 24,947.75 kWh |

### Demand breakdown

| Category | Peak (kW) | Daily (kWh) | Share | Priority |
|---|---|---|---|---|
| Essential services | 5.44 | 39.09 | 34.7% | Non-negotiable — protected under all scenarios |
| Productive activities | 9.66 | 29.26 | 61.7% | High — drives economic value and ROI |

### Capacity planning guidance

| | |
|---|---|
| **Recommended capacity** | 19.57 kW (includes 25% headroom for growth and contingency) |
| **Critical load reserve** | 25% baseline reserved for essential services (architecturally enforced) |
| **Minimum transformer sizing** | 24.47 kVA (assuming 0.8 power factor) |
| **Distribution voltage** | 11 kV or 33 kV recommended for productive-use loads |
| **Service connections (estimated)** | ≈ 10 productive-use connection points |

**Bankability disclaimer.** All figures are conservative lower-bound estimates intended to support institutional planning. Actual demand may be 20–40% higher than presented; peak demand may reach 1.5× the conservative estimate during high-coordination periods. Infrastructure should be sized with growth headroom and modular expansion capability, and this document does not constitute an engineering design or financing approval.

# 6 Risk, Uncertainty & Governance

### Demand uncertainty

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|---|---|
| **Conservative estimate** | Lower-bound, as presented in Section 5 |
| **Expected range** | Actual demand likely 20–40% higher than conservative estimate |
| **Upper bound** | Peak demand may reach 1.5× conservative estimate during high-coordination periods |
| **Mitigation** | 25% headroom in sizing; modular expansion capability |

### Coordination persistence risk

Coordination patterns may weaken or shift if economic, agronomic, or seasonal conditions change. At the close of the seven-cycle window, two patterns demonstrated high stability (≥ 5 of 7 cycles, strongly validated by telemetry).

- **Continuous monitoring** — re-evaluate coordination patterns every 4–8 weeks.
- **Adaptive capacity** — design infrastructure for flexible load allocation.
- **Stakeholder engagement** — maintain communication with productive-use actors.
- **Phased deployment** — start with high-confidence zones and expand as patterns persist.

### Governance principles

1. Essential services receive non-negotiable priority (25% baseline reserve).
2. Productive-use activities are allocated based on coordination confidence.
3. Remaining capacity is available for household and commercial use.
4. Emergency scenarios may scale essential capacity to 50%.

### Monitoring cadence

| | |
|---|---|
| **Re-evaluation frequency** | Every 4–8 weeks |
| **Adaptive management** | Adjust capacity allocation based on observed patterns and community feedback |
| **Audit basis** | All outputs auditable against KULIMA OS system invariants (AGENTS.md) |

# 7 Deployment Readiness Snapshot

| Dimension | Status | Notes |
|---|---|---|
| Technical | HIGH | Demand signals validated; load estimates conservative; requirements clear |
| Financial | MODERATE | Requires DFI commitment, tariff approval, and cost-recovery plan |
| Institutional | MODERATE | Requires utility engagement, regulatory approval, governance framework |
| Community | LOW | Stakeholder engagement not yet initiated; required before deployment |
| Overall | MODERATE | Technical foundation strong; institutional and community engagement needed |

### Indicative implementation timeline

| | |
|---|---|
| **Phase 1 — Planning** | 3–6 months |
| **Phase 2 — Construction** | 6–12 months |
| **Phase 3 — Operation** | Ongoing |
| **Total timeline** | 9–18 months from approval to full operation |

### Indicative CAPEX (lower-bound)

| | |
|---|---|
| **Transformer & equipment** | USD 3,670 (≈ USD 150 / kVA) |
| **Service connections** | USD 5,000 (≈ USD 500 per connection) |
| **Contingency** | Add 20–30% for unforeseen costs |

### Next institutional steps

1. Secure financing commitment from a DFI or development finance institution.
2. Engage utility operator to confirm grid connection point and O&M; responsibility.
3. Initiate community stakeholder engagement and consultation.
4. Commission detailed engineering design and ESIA.
5. Obtain regulatory approvals (tariff, safety, environmental).
6. Procure equipment and select construction contractor.
7. Construct infrastructure with community liaison and safety protocols.
8. Commission and activate service connections.
9. Monitor demand realisation and adjust capacity allocation.
10. Establish ongoing monitoring and evaluation framework.

## 8 Ethics & Data Governance

KULIMA OS is designed as Digital Public Infrastructure with privacy and non-surveillance as structural properties of the system, not policy overlays. The following invariants apply to every output, including this prospectus.

| | |
|---|---|
| **Zero-PII guarantee** | No personal identifiers appear in any signal, intermediate output, or published artifact |
| **No surveillance or profiling** | No credit scoring, individual profiling, or real-time tracking is performed |
| **Coordination over identity** | The system reasons over collective patterns only; identity is never required or inferred |
| **Temporal moat** | All processing occurs in time-batched windows; no real-time observation of individuals |
| **Critical load protection** | Essential services receive non-negotiable, architecturally enforced priority |
| **Data minimisation** | Raw signals are never stored or exported; only aggregated patterns cross institutional boundary |
| **Auditability** | All outputs are auditable against the KULIMA OS system invariants (AGENTS.md) |

### Pilot scope and limitations

• **Demonstrated:** coordination-pattern detection from identity-free signals; pattern-based trust without participant reputation; architectural protection of essential services; institutional-grade outputs for infrastructure planning.
• **Simulated:** telemetry validation (synthetic telemetry); essential-service signals (telemetry only, no human coordination signals in pilot); spatial zoning (manually specified).
• **Excluded:** real SMS/WhatsApp transport; steward review interface (auto-approved in pilot); multi-cycle longitudinal tracking beyond the seven-cycle window; real-world hardware integration.

> **Sample notice.** This is a pilot demonstration using synthetic signals. Production deployment requires real signal ingestion, steward review, and community engagement. This document does not constitute a financing approval, engineering design, or guarantee of future demand.

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